

TECHNICAL REQUIREMENT DOCUMENT (TRD)

Project Name

Product Based Tree View Send & Receive Helping Plan Management System

Version

1.0

Document Type

Technical Requirement Document (TRD)

Technology Stack

Backend

- Python 3.12+
- Django 5.x
- Django REST Framework (DRF)

Frontend

- HTML5
- CSS3
- Bootstrap 5
- JavaScript
- jQuery / AJAX

Database

- PostgreSQL 16+

File Storage

- Local Storage (Development)

- AWS S3 / DigitalOcean Spaces (Production)

Web Server

- Nginx

Application Server

- Gunicorn

Operating System

- Ubuntu 24.04 LTS

Version Control

- Git
 - GitHub / GitLab
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SYSTEM ARCHITECTURE

User Roles

Super Admin

Full System Access

Admin

Operational Access

Member/User

Limited User Access

DJANGO APPLICATION STRUCTURE

Core Apps

accounts

Purpose:

- Registration
- Login
- User Profile
- Password Management

dashboard

Purpose:

- Dashboard Statistics
- Analytics

network

Purpose:

- Referrals
- Binary Tree
- Left Team
- Right Team

kyc

Purpose:

- KYC Submission
- Verification Workflow

pmf

Purpose:

- PMF Payment
- PMF History
- PMF Approval

helpdesk

Purpose:

- Send Help
- Receive Help

rewards

Purpose:

- Reward Management

support

Purpose:

- Ticket Management

notifications

Purpose:

- Notifications
- QR Code Management

products

Purpose:

- Product Management

reports

Purpose:

- Reporting Module

DATABASE DESIGN

users

Field	Type
id	BIGINT

member_id	VARCHAR(20)
sponsor_id	VARCHAR(20)
username	VARCHAR(100)
email	VARCHAR(150)
mobile	VARCHAR(20)
password	HASH
status	BOOLEAN
created_at	DATETIME

user_profiles

Field	Type
id	BIGINT
user_id	FK
address	TEXT
city	VARCHAR
state	VARCHAR
country	VARCHAR
pincode	VARCHAR
profile_image	IMAGE

referrals

Field	Type
id	BIGINT
sponsor_id	FK
member_id	FK

position LEFT/RIGHT

joining_date DATETIME

kyc_requests

Field	Type
id	BIGINT
user_id	FK
aadhaar_number	VARCHAR
pan_number	VARCHAR
aadhaar_file	FILE
pan_file	FILE
bank_file	FILE
photo_file	FILE
status	PENDING/APPROVED/REJECTED
remarks	TEXT

pmf_requests

Field	Type
id	BIGINT
user_id	FK
amount	DECIMAL
payment_proof	FILE
status	PENDING/APPROVED/REJECTED
created_at	DATETIME

send_help_requests

Field	Type
id	BIGINT
user_id	FK
amount	DECIMAL
payment_method	VARCHAR
proof_file	FILE
status	PENDING/APPROVED/REJECTED

receive_help_requests

Field	Type
id	BIGINT
user_id	FK
amount	DECIMAL
status	PENDING/APPROVED/REJECTED

rewards

Field	Type
id	BIGINT
user_id	FK
reward_name	VARCHAR
reward_type	PRODUCT/CASH
qualification_date	DATE
status	PENDING/APPROVED/REJECTED

support_tickets

Field	Type
id	BIGINT
user_id	FK
subject	VARCHAR
message	TEXT
attachment	FILE
status	OPEN/PENDING/CLOSED

notifications

Field	Type
id	BIGINT
title	VARCHAR
description	TEXT
qr_image	IMAGE
status	BOOLEAN

products

Field	Type
id	BIGINT
product_code	VARCHAR
product_name	VARCHAR
description	TEXT

image	IMAGE
price	DECIMAL
status	BOOLEAN

ADMIN PANEL REQUIREMENTS

Dashboard

Statistics Required:

- Active Members
- Inactive Members
- Total Members
- Today's Joining
- Today's Activation
- Total Help Provided
- Pending Help Requests
- Accepted Help Requests
- Rejected Help Requests
- Pending PMF Requests

API:

GET /api/admin/dashboard/

Associate Management

Features:

- List Members
- Search Members
- Activate Member
- Deactivate Member
- View Profile

APIs:

GET /api/admin/members/

POST /api/admin/member/activate/

POST /api/admin/member/deactivate/

Tree View

Features:

- Binary Tree View
- Search Member
- View Downline

API:

GET /api/network/tree/{member_id}/

KYC Management

Features:

- Pending KYC
- Approved KYC
- Rejected KYC

APIs:

GET /api/admin/kyc/

POST /api/admin/kyc/approve/

POST /api/admin/kyc/reject/

PMF Request Management

Features:

- View Requests
- Approve PMF
- Reject PMF

APIs:

GET /api/admin/pmf/

POST /api/admin/pmf/approve/

POST /api/admin/pmf/reject/

Product Dispatch

Features:

- Generate Dispatch
- Tracking Number
- Dispatch History

Tables:

product_dispatch

dispatch_history

USER PANEL REQUIREMENTS

Dashboard

Display:

- Member ID
- Sponsor ID
- Account Status
- KYC Status

Quick Actions:

- Top Achievers
- My Profile
- KYC
- PMF

- Send Help
 - Receive Help
 - My Referrals
 - Password
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Profile Management

Features:

- View Profile
- Update Profile
- Upload Profile Image

API:

GET /api/profile/

PUT /api/profile/

Network Module

Features:

- Direct Referrals
- Left Team
- Right Team
- Tree View

API:

GET /api/network/referrals/

GET /api/network/tree/

PMF Module

Features:

- Pay PMF
- PMF History
- PMF Help Table

API:

POST /api/pmf/pay/

GET /api/pmf/history/

Send Help Module

API:

POST /api/send-help/

GET /api/send-help/history/

Receive Help Module

API:

POST /api/receive-help/

GET /api/receive-help/history/

Reward Module

API:

GET /api/rewards/

Support Module

API:

POST /api/support/create/

GET /api/support/list/

SECURITY REQUIREMENTS

Authentication:

- JWT Token Authentication
- Refresh Tokens

Authorization:

- Role Based Access Control (RBAC)

Password:

- PBKDF2 Password Hashing

Protection:

- CSRF Protection
 - XSS Protection
 - SQL Injection Protection
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REPORTS

Admin Reports:

- Member Report
- KYC Report
- PMF Report
- Product Report
- Reward Report
- Support Report

User Reports:

- Referral Report

- Team Report
- PMF Report
- Send Help Report
- Receive Help Report

Export Formats:

- Excel
 - PDF
 - CSV
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PERFORMANCE REQUIREMENTS

Dashboard Load Time:
< 3 Seconds

API Response:
< 500 ms

Concurrent Users:
5,000+

Database Records:
1,000,000+

DEPLOYMENT REQUIREMENTS

Production Server:

CPU: 8 Core

RAM: 16 GB

Storage: 500 GB SSD

Database: PostgreSQL

Web Server: Nginx

Application Server: Gunicorn

SSL: Required

Daily Backup: Required

Monitoring: Required

Uptime Target: 99.9%